

VISIT · VICTR, VANDERBILT UNIVERSITY MEDICAL CENTER · 8 OCTOBER 2026

From an EHR Form to SDTM

JASPEHR, FHIR and CDISC — a year on



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National Project on Data Standardization

Japan AMED-funded project (2025–2027)

“Implementation Research on Standardized RWD Infrastructure for Post-Marketing Safety Measures through FHIR, CDISC, and SS-MIX2 Integration.”

- Creating a seamless pipeline between healthcare data (**FHIR** format) and regulatory research data (**CDISC** format)
- Leveraging Japan's existing **SS-MIX2** standard for EHR data exchange
- Establishing national standards for real-world data (RWD) in post-marketing safety surveillance
- Developing a framework for automated data transformation between clinical care and research



I serve as the Principal Investigator of this project

Leading CDISC adoption in academia through three consecutive AMED-funded projects

2019–2021

Phase 1

2022–2024

Phase 2

2025–2027

Current phase

Japan's PMS landscape — and our scope

EHR → CDISC SDTM, directly, for PMS use.

- ✓ **EHR penetration** — high; FHIR R4 / SS-MIX2 widely adopted
- ✓ **Regulatory PMS frameworks** (GPSP, re-examination, re-evaluation) generate substantial RWD obligations
- ✗ **PMS data collection is still mostly proprietary**, not natively in CDISC standards



Arithmetic of why input-time encoding matters

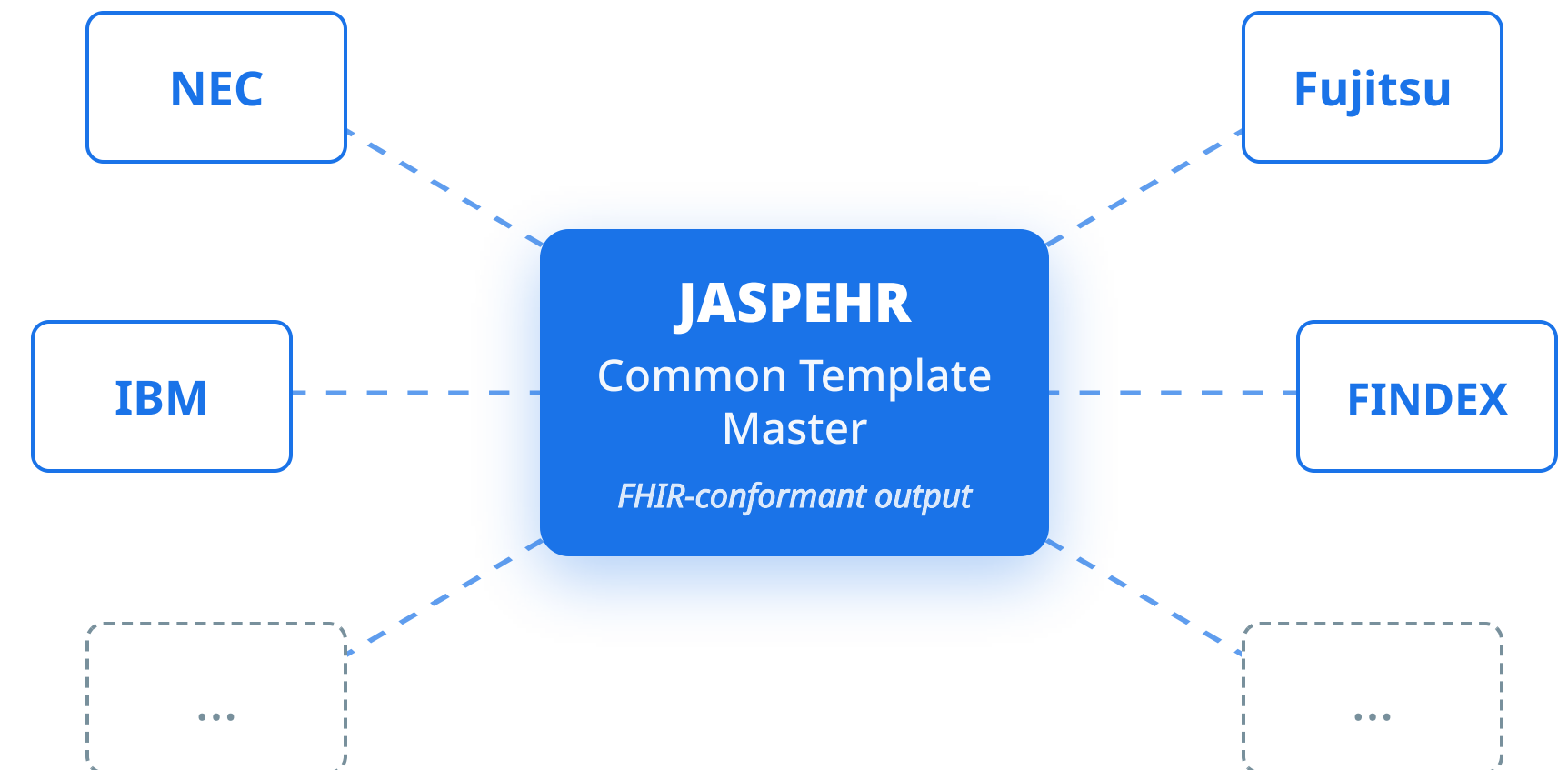
Mapping after the fact $\approx 2,000$ vars $\times$ 1,000,000 records = **2 billion decisions per study.**

Encoding at input $\approx$ **2,000 template-design decisions, once.**

JASPEHR in one slide

Japanese Standard Platform for EHRs — AMED-funded, launched by Prof. Kengo Miyo at JIHS (formerly NCGM).

- **Vendor-independent EHR template spec** — distributed as a "common template master"
- Each EHR vendor **auto-converts** it into its own template engine
- **FHIR-native** — based on Questionnaire / QuestionnaireResponse
- **In active use** across multiple Japanese research projects



This is the platform our CDISC extension builds on.

Embedding CDISC into FHIR — four metadata

Goal: make EHR input natively CDISC-conformant.

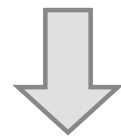
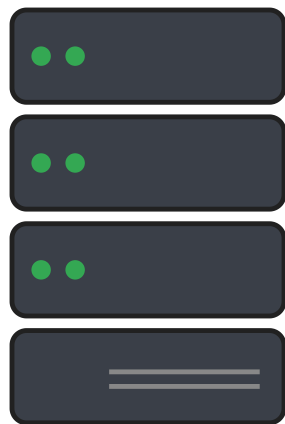
#	CDISC metadata	Scale	FHIR realisation	Status
1	SDTM variable names	~70 domains · 2–3K variables	item.code on Questionnaire	✓ JSON sample delivered
2	Controlled Terminology	1,181 ValueSets · ~50K choices	FHIR CodeSystem + ValueSet	CDISC chose persistent IDs (2 Oct 2026)
3	MedDRA	90K LLT terms · 48.6 MB	FHIR CodeSystem	✓ JSON sample delivered
4	IDF, WHODrug WHODrug optional — sites without a licence stop at IDF	IDF (JP drug master) + WHODrug 30K terms · 5.7 MB	FHIR CodeSystem (IDF-linked)	✓ JSON sample delivered

Verification environment

How we author and validate CDISC-extended FHIR Questionnaires before vendor hand-off.

1. NLM Form Builder

localhost:9030 · BSD



Create & export
FHIR Questionnaire

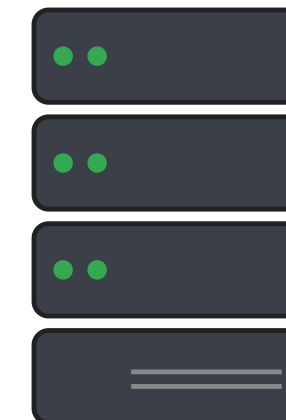
Set answer choices



Returns matching metadata

2. HAPI FHIR Server

localhost:8080 · Apache 2.0



Load



- SDTM variables
- CT
- MedDRA
- IDF, WHODrug

*Autocomplete responsiveness validated
against MedDRA (90K) and WHODrug (30K).*

— Developer's local environment